systems, they offer the advantage of local execution, improved privacy, and independence from network connectivity.

What is LangChain?

LangChain is an open source framework designed for building AI agents, orchestrating workflows and experimenting with AI agents, agent architectures and agent orchestration. It enables developers to build intelligent systems without being tied to a specific language model.

One of its biggest advantages is its standardised model interface. Different language models, such as GPT, Gemma and others, expose different APIs. The abstraction offered by LangChain is that of providing a common interface to help developers easily switch between models.

Beyond providing a standardised interface, LangChain enables developers to build and orchestrate AI agents while experimenting with different agent architectures. It also supports data compression within agent workflows and provides observability, debugging and monitoring capabilities, providing greater flexibility for experimentation with different AI models.

With such characteristics, developers can try out increasingly advanced applications of AI without losing flexibility in relation to various models.

Why use LangChain?

One of LangChain's biggest strengths is its ability to provide a standardised interface across multiple large language models. Whether developers are working with Gemini, ChatGPT, Claude or other models, LangChain abstracts their different APIs into a common framework, simplifying agent orchestration. This framework also allows for debugging, observability and monitoring capabilities.

Understanding the LangChain ecosystem

LangChain is just one component of an entire toolchain ecosystem where every tool performs a specific function within the context of AI application development.

LangChain: LangChain provides the entry point for building AI agents. It is suitable for developing straightforward applications such as image recognition, reasoning tasks and conversational agents without requiring complex orchestration.

LangGraph: The more sophisticated the application, the more powerful are the orchestration capabilities provided by LangGraph. This is because LangGraph enables multiple agents to coordinate tasks, manage workflows, and execute advanced agent orchestration.

Deep Agents: Deep Agents provide developers with lower-level control over AI systems. They are designed for scenarios requiring hardware optimisation.

LangSmith: Unlike the other components, LangSmith focuses on observability rather than agent execution. This includes monitoring, debugging, and visualisation of workflow, enabling developers to monitor and debug deployed agents.

Building an edge AI agent with Gemma

A practical example combines LangChain with Google's Gemma models for building an AI application that can understand images and produce brief descriptions. It starts when an image is uploaded on the edge device by the user. An RGB conversion of the image takes place before it is analysed by PaliGemma, which is a vision-language model that has around 3 billion parameters. PaliGemma generates an image caption describing the uploaded image.

Once the image caption has been generated, a LangChain agent processes the output by invoking two separate tools. The first tool uses PaliGemma to generate an image caption, while the second uses the Gemma 2B model to generate a summary based on that caption. The agent then combines both outputs into a structured response containing the image caption and its corresponding summary.

Application workflow

The demonstration employs two lightweight AI models. PaliGemma (3B) is used to generate image captions, while Gemma 2B performs text summarisation based on those captions. Together, the models occupy approximately 7-8GB of storage, allowing the complete application to run locally on a standard laptop equipped with around 16GB RAM.

A Gradio web interface provides a simple front-end where users upload images and receive generated captions alongside AI-generated summaries. LangChain manages the communication of both models, enabling image captioning and text summarisation to be executed as a LangChain agent pipeline.

Rapid prototyping with Google Colab

Google Colab provides a convenient environment for experimenting with edge AI applications. Developers are allowed to choose hardware accelerators like NVIDIA T4 GPUs to accelerate the inference of their applications.

The workflow begins by installing the required libraries, including Transformers, LangChain, PyTorch, and other supporting packages. After installation, the runtime is rebooted and then the open source models are downloaded from the Hugging Face platform, which requires user authentication. After the models have been downloaded, LangChain components such as prompt templates, tools, agents, and language model chains are imported to begin building the application.

Loading lightweight Gemma models

The application uses PaliGemma 3B, a vision-language model designed for image captioning. Running locally, the model is configured to execute on